

Introduction to The Cash in Advance Model

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I Introduction

Consider a dynamic cash in advance model with an infinite time horizon (indexed by $t = 0, 1, 2, 3, \dots$). At the beginning of a period, the consumer has M_{t-1}^d units of money, B_{t-1} nominal bonds, and f_{t-1} real bonds. A nominal bond issued in $t - 1$ pays one unit of money at the beginning of t . Holding B_{t-1} units of nominal bonds yields B_{t-1} units of money in t . A real bond issued in $t - 1$ pays one unit of consumption goods at the beginning of t . Holding f_{t-1} units of real bonds yields f_{t-1} units of goods in t . Finally, M_{t-1}^d units of money is exchanged for M_{t-1}^d units of money in the next period. The government levies lump sum taxes, τ_t , that are rebated back to the consumer. Then, the asset market opens and the consumer can use their beginning of period wealth to purchase bonds and obtain money in order to purchase consumption goods. For clarity, we express all asset quantities in terms of consumption goods. So, letting $\pi_t = \frac{P_t}{P_{t-1}} - 1$ be the inflation rate and lower case variables denote the real quantities we can obtain the asset market constraint

$$c_t + q_t b_t + s_t f_t \leq \frac{m_{t-1}^d}{1 + \pi_t} + \frac{b_{t-1}}{1 + \pi_t} + f_{t-1} + \tau_t \quad (1)$$

In other words, consumption plus the purchase of bonds must be less than or equal to the consumers existing money, income from previous periods bonds, and the lump sum rebate from the government. Once the asset market closes, the consumer goes to work and supplies labor. Note that their wage will not be paid until the end of the period (after the firm sells its output). Before wages from today's work are paid, the goods market opens, and the consumer must use the money they have on hand to purchase consumption goods. Thus, the real quantity of money is equal to the quantity of consumption goods. Once the firm pays wages, the consumer carries their assets into the next period. This yields the budget constraint:

$$c_t + q_t b_t + s_t f_t + m_t^d \leq \frac{m_{t-1}^d}{1 + \pi_t} + \frac{b_{t-1}}{1 + \pi_t} + f_{t-1} + \tau_t + w_t n_t^s \quad (2)$$

[†]Definitions and example problems from ECON 62051: Macroeconomic Theory I instructed by Dr. Duhyeong Kim and *Notes on Cash in Advance* by Steve Williamson. Kent State University.

The consumers preferences are

$$U\left(\{c_t, n_t^s\}_{t=0}^{\infty}\right) = \sum_{t=0}^{\infty} \beta^t \left(u(c_t) - v(n_t^s)\right)$$

Where $\beta \in (0, 1)$ is the discount factor¹. Note that u is continuously differentiable, strictly increasing and strictly concave. The Inada conditions are

$$\lim_{c \rightarrow 0} u'(c) = \infty, \quad \lim_{c \rightarrow \infty} u'(c) = 0$$

$v(n_t^s)$ is the disutility of labor, where v is continuously differentiable, strictly increasing, and strictly convex. Note $0 \leq n_t^s \leq \bar{n}$ The Inada conditions are:

$$\lim_{n \rightarrow 0} v'(n) = 0, \quad \lim_{n \rightarrow \bar{n}} v'(n) = \infty$$

The representative firm has a linear production technology:

$$y_t = a n_t^d \tag{3}$$

Where $a > 0$ is productivity. It is how much output each unit of labor will produce. Solving the firm's problem:

$$\begin{aligned} & \max(an_t^d - wn_t^d) \\ & \frac{\partial}{\partial n_t^d} = a - w = 0 \end{aligned}$$

Here w_t depends on a so we have to make cases depending on if w_t is less than, greater than, or equal to w_t . So, in equilibrium $w_t = a \quad \forall t \geq 0$, and the firms labor demand is

$$n_t^d = \begin{cases} 0 & \text{if } w_t > a \\ \in [0, \infty) & \text{if } w_t = a \\ \infty & \text{if } w_t < a \end{cases}$$

Assuming $B_t = G_t = 0$ & $\tau_t = -\frac{T_t}{P_t}$ The government's budget constraint is (in terms of units of goods):

$$\tau_t = m_t^s - \frac{m_{t-1}^s}{1 + \pi_t} \tag{4}$$

¹where $\beta = \frac{1}{(1+r)^t}$

II Solving the Consumers Problem

Given $\{w_t, q_t, s_t, \pi_{t+1}\}_{t=0}^{\infty}$, the consumer solves:

$$\max_{\{c_t, n_t^s, m_t^d, b_t, f_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t [u(c_t) - v(n_t^s)]$$

s.t.

$$(1) \quad c_t + q_t b_t + s_t f_t \leq \frac{m_{t-1}^d}{1 + \pi_t} + \frac{b_{t-1}}{1 + \pi_t} + f_{t-1} + \tau_t$$

$$(2) \quad c_t + q_t b_t + s_t f_t + m_t^d \leq \frac{m_{t-1}^d}{1 + \pi_t} + \frac{b_{t-1}}{1 + \pi_t} + f_{t-1} + w_t n_t^s + \tau_t$$

To solve this, we must use a bellman equation. Let $\psi_t = (m_t^d, b_t, f_t)$ be the vector of money and asset holdings and let $\varphi_t = (w_t, q_t, s_t, \pi_t)$ be the vector of prices and the inflation rate. Finally, let $W(\psi_{t-1}, \varphi_t, \tau_t)$ be the maximum utility of the consumer from now onward, given $(\psi_{t-1}, \varphi_t, \tau_t)$. Then, the bellman equation (or the recursive formulation) is

$$W(\psi_{t-1}, \varphi_t, \tau_t) = \max_{\{c_t, \psi_t, n_t^s\}_{t=0}^{\infty}} \left(u(c_t) - v(n_t^s) + \beta W(\psi_t, \varphi_{t+1}, \tau_{t+1}) \right)$$

s.t.

$$(1) \quad c_t + q_t b_t + s_t f_t \leq \frac{m_{t-1}^d}{1 + \pi_t} + \frac{b_{t-1}}{1 + \pi_t} + f_{t-1} + \tau_t$$

$$(2) \quad c_t + q_t b_t + s_t f_t + m_t^d \leq \frac{m_{t-1}^d}{1 + \pi_t} + \frac{b_{t-1}}{1 + \pi_t} + f_{t-1} + w_t n_t^s + \tau_t$$

Where $\beta W(\psi_t, \varphi_{t+1}, \tau_{t+1})$ represents the discounted value of the assets brought to next period (which will change the value function). The first step to solving this dynamic programming problem is to solve the maximization problem on the RHS. Let λ_t, μ_t be the Lagrange multipliers. Then, the Lagrangian function is

$$\begin{aligned} \mathcal{L}_t = & u(c_t) - v(n_t^s) + \beta W(\psi_t, \varphi_{t+1}, \tau_{t+1}) + \lambda_t \left(\frac{m_{t-1}^d}{1 + \pi_t} + \frac{b_{t-1}}{1 + \pi_t} + f_{t-1} + \tau_t - c_t - q_t b_t - s_t f_t \right) \\ & + \mu_t \left(\frac{m_{t-1}^d}{1 + \pi_t} + \frac{b_{t-1}}{1 + \pi_t} + f_{t-1} + w_t n_t^s + \tau_t - c_t - q_t b_t - s_t f_t - m_t^d \right) \end{aligned}$$

Obtaining the FOC's:

$$\begin{aligned}
\frac{\partial \mathcal{L}_t}{\partial c_t} &= u'(c_t) - \lambda_t - \mu_t = 0, \\
\frac{\partial \mathcal{L}_t}{\partial n_t^s} &= -v'(n_t^s) + \mu_t w_t = 0, \\
\frac{\partial \mathcal{L}_t}{\partial m_t^d} &= \frac{\partial \beta W(\psi_t, \varphi_{t+1}, \tau_{t+1})}{\partial m_t^d} - \mu_t = 0, \\
\frac{\partial \mathcal{L}_t}{\partial b_t} &= \frac{\partial \beta W(\psi_t, \varphi_{t+1}, \tau_{t+1})}{\partial b_t} - \lambda_t q_t - \mu_t q_t = 0, \\
\frac{\partial \mathcal{L}_t}{\partial f_t} &= \frac{\partial \beta W(\psi_t, \varphi_{t+1}, \tau_{t+1})}{\partial f_t} - \mu_t s_t - \lambda_t s_t = 0
\end{aligned}$$

Each F.O.C. that contains the partial derivative of W cannot directly be solved like our consumption or labor F.O.Cs. So, we differentiate both sides of the bellman equation (equivalently we take the Lagrangian with respect to the variables in $t - 1$ using the Envelope Theorem). Our goal is to then index those derivatives up one period and substitute back into our F.O.Cs to rid of the nasty partial derivatives of W . In doing this, we can apply the Envelope Theorem to rid of any indirect effects and obtain simple equations to plug back in.

$$\begin{aligned}
\frac{\partial W(\psi_{t-1}, \varphi_t, \tau_t)}{\partial m_{t-1}^d} &= \frac{\partial \mathcal{L}_t}{\partial m_{t-1}^d} = \frac{\lambda_t + \mu_t}{1 + \pi_t} \\
\frac{\partial W(\psi_{t-1}, \varphi_t, \tau_t)}{\partial b_{t-1}^d} &= \frac{\partial \mathcal{L}_t}{\partial b_{t-1}^d} = \frac{\lambda_t + \mu_t}{1 + \pi_t} \\
\frac{\partial W(\psi_{t-1}, \varphi_t, \tau_t)}{\partial f_{t-1}^d} &= \frac{\partial \mathcal{L}_t}{\partial f_{t-1}^d} = \lambda_t + \mu_t
\end{aligned}$$

Updating by one period and substituting back into the F.O.Cs gives us

$$\begin{aligned}
\frac{\partial \mathcal{L}_t}{\partial c_t} &= u'(c_t) - \lambda_t - \mu_t = 0, \\
\frac{\partial \mathcal{L}_t}{\partial n_t^s} &= -v'(n_t^s) + \mu_t w_t = 0, \\
\frac{\partial \mathcal{L}_t}{\partial m_t^d} &= \beta \frac{\lambda_{t+1} + \mu_{t+1}}{1 + \pi_{t+1}} - \mu_t = 0, \\
\frac{\partial \mathcal{L}_t}{\partial b_t} &= \beta \frac{\lambda_{t+1} + \mu_{t+1}}{1 + \pi_{t+1}} - \lambda_t q_t - \mu_t q_t = 0, \\
\frac{\partial \mathcal{L}_t}{\partial f_t} &= \beta (\lambda_{t+1} + \mu_{t+1}) - \mu_t s_t - \lambda_t s_t = 0
\end{aligned}$$

Then, we can solve for λ_t, μ_t in the first two F.O.Cs in order to re write the rest:

$$\begin{aligned}
u'(c_t) &= \lambda_t + \mu_t \\
\mu_t &= \frac{v'(n_t^s)}{w_t}
\end{aligned}$$

Solving for the rest of the F.O.Cs:

$$\begin{aligned}
\lambda_t + \mu_t &= u'(c_t) \\
\mu_t &= \frac{v'(n_t^s)}{w_t} \\
v'(n_t^s) &= \frac{\beta w_t u'(c_{t+1})}{1 + \pi_{t+1}} \\
q_t u'(c_t) &= \frac{\beta u'(c_{t+1})}{1 + \pi_{t+1}} \\
s_t u'(c_t) &= \beta u'(c_{t+1})
\end{aligned}$$

Note that we have 7 variables but 5 equations, so we must add in the budget constraints. Thus, the KKT conditions are:

$$\begin{aligned}
\text{i.} \quad & \lambda_t + \mu_t = u'(c_t) \\
\text{ii.} \quad & \mu_t = \frac{v'(n_t^s)}{w_t} \\
\text{iii.} \quad & v'(n_t^s) = \frac{\beta w_t u'(c_{t+1})}{1 + \pi_{t+1}} \\
\text{iv.} \quad & q_t u'(c_t) = \frac{\beta u'(c_{t+1})}{1 + \pi_{t+1}} \\
\text{v.} \quad & s_t u'(c_t) = \beta u'(c_{t+1}) \\
\text{vi.} \quad & \lambda_t \left(\frac{m_{t-1}^d}{1 + \pi_t} + \frac{b_{t-1}}{1 + \pi_t} + f_{t-1} + \tau_t - c_t - q_t b_t - s_t f_t \right) = 0 \\
\text{vii.} \quad & \mu_t \left(\frac{m_{t-1}^d}{1 + \pi_t} + \frac{b_{t-1}}{1 + \pi_t} + f_{t-1} + w_t n_t^s + \tau_t - c_t - q_t b_t - s_t f_t - m_t^d \right) = 0
\end{aligned}$$

There are some F.O.Cs of interest. The third says that the marginal disutility of supplying labor is equal to the discounted effective real wage times the marginal utility of consumption in the next period. The fourth says that to buy a nominal bond, the consumer gives up q_t/P_t units of consumption that is valued at $u'(c_t)$. For the real bond F.O.C, the consumer gives up s_t units of consumption, which is again valued at $u'(c_t)$.

III Competitive Equilibrium

a. Definition

Given a, i, π_0 , and M_0 , a competitive equilibrium is a set of quantities $\{c_t, n_t^s, n_t^d, y_t, m_t^s, m_t^d, b_t, f_t, \tau_t\}_{t=0}^\infty$ and prices $\{\pi_{t+1}, q_t, s_t, w_t\}_{t=0}^\infty$ such that:

1. The consumer chooses $\{c_t, n_t^s, m_t^d, b_t, f_t\}_{t=0}^\infty$ given the prices and τ_t .
2. Firm chooses $\{y_t, n_t^d\}_{t=0}^\infty$ given $\{w_t\}_{t=0}^\infty$ and a .
3. Government sets $\{\tau_t\}_{t=0}^\infty$ to satisfy its budget constraint given i, M_0 .

Note that for the consumer, to remember what to include, we must match prices to the labor and bond variables, match inflation to money demanded, and taxes to consumption. Also note that, in problem set 7, y_t is exogenous since there is no labor. Thus, the firm cannot choose it.

IV Government Policy

In the model, we assume that the government sets transfers over time such that the nominal money supply grows at constant rate i . Note that in assignment 7 it was i_t which is not constant. Back to the constant growth rate model we can say that

$$M_{t+1} = (1 + i)M_t \quad \forall t \geq 0$$

This implies

$$M_t = (1 + i)^t M_0 \quad \& \quad 1 + \pi_{t+1} = (1 + i) \frac{m_t}{m_{t+1}}$$

a. Equilibrium Conditions

To solve for the equilibrium conditions, we first let $n_t^s = n_t^d = n_t$ and $m_t^s = m_t^d = m_t$. Then, we can use two constraints for a shortcut (assuming that they are binding) the important one is subtracting the asset market constraint from the budget constraint. This also is what we did in Assignment 7 to get the equilibrium conditions for the price of the financial instruments there. In this model, the subtraction yields $m_t^d = w_t n_t^s$. So, $m_t = y_t = c_t = a n_t$. Also recall that $m_t = P_t y_t$ so then the quantity theory of money $MV = PY$ implies that $V = 1$ in our model.

The second constraint we can use for a shortcut is $v'(n_t^s) = \frac{\beta w_t u'(c_{t+1})}{1 + \pi_{t+1}}$. In equilibrium, we can use that $w_t = a$, $m_t = y_t = a n_t = c_t$, and $1 + \pi_{t+1} = (1 + i) \frac{m_t}{m_{t+1}}$ to obtain

$$\begin{aligned} v'(n_t^s) &= \frac{\beta w_t u'(c_{t+1})}{1 + \pi_{t+1}} \implies v'(n_t) = \frac{\beta a u'(a n_{t+1})}{1 + i \frac{m_t}{m_{t+1}}} \implies \\ v'(n_t) &= \frac{\beta a u'(a n_{t+1}) a n_{t+1}}{(1 + i) a n_t} \implies n_t v'(n_t) = \frac{a n_{t+1}}{1 + i} \beta u'(a n_{t+1}) \end{aligned}$$

This is the first order non-linear difference equation that determines the path of n_t . This general trend follows for all of the F.O.Cs. I suggest listing out that $w_t = a$, $m_t = y_t = an_t = c_t$, and $1 + \pi_{t+1} = (1 + i)^{\frac{m_t}{m_{t+1}}}$ and just boiling down as much as you can to n_t and i to get these quantities. In a problem with no labor, use y_t like in Assignment 7.

V Steady State

For the steady state, set $n_t = n \forall t \geq 0$. Then it will all cancel nicely to

- i. $v'(n) = \frac{a\beta u'(an)}{1+i}$
- ii. $m = c = y = an$
- iii. $w = a$
- iv. $q = \frac{\beta}{1+i}$
- v. $s = \beta$
- vi. $\pi = i$
- vii. $M_t = (1+i)^t M_0$
- viii. $P_t = \frac{(1+i)^t M_0}{an}$

Note that money is **neutral** if the changes in money supply affect nominal variables (like the capital price levels) but not real variables (lower case ones). In this model, M_0 only appears in the M_t and P_t variables so money *is* neutral. Money is **super neutral** if the growth rate of the money supply, i , affects nominal variables but not real ones. In this model money *is not* super neutral since our condition $v'(n) = \frac{a\beta u'(an)}{1+i}$ depends on i , and since $c = y = an$ then it affects consumption and output too.

Also, note that we can find the nominal interest rate as $R = \frac{1}{q} - 1$ and the real interest rate as $r = \frac{1}{s} - 1$. This relates to the Fisher effect that says $R = r + \pi$. The Fisher Effect also states that a permanent 1 percent increase in the money growth rate results in a 1 percent permanent increase in the inflation rate and the nominal interest rate. Finally, we can interpret the first equilibrium condition in the steady state. It says that the marginal disutility of supply labor is equal to the effective real wage, $\frac{a}{1+i}$, times the discounted marginal payoff for the consumer, $\beta u'(an)$. So, income today is spent tomorrow, implying that inflation works like a tax. In other words, high inflation lowers the real value of today's income. This is a good representation of the long run effects, but in the short run central banks say that monetary expansion reduces nominal interest rates and increases inflation.

VI Pareto Optimal Allocations